

Soddy Circles, Inversion, and Descartes' Circle Theorem

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The Kiss Precise (1936)

Frederick Soddy

*For pairs of lips to kiss maybe
Involves no trigonometry.
This not so when four circles kiss
Each one the other three.
To bring this off the four must be
As three in one or one in three.
If one in three, beyond a doubt
Each gets three kisses from without.
If three in one, then is that one
Thrice kissed internally.*

*Four circles to the kissing come.
The smaller are the benter.
The bend is just the inverse of
The distance from the center.
Though their intrigue left Euclid dumb
There's now no need for rule of thumb.
Since zero bend's a dead straight line
And concave bends have minus sign,
The sum of the squares of all four bends
Is half the square of their sum.*

1 What are Soddy Circles?

When given three mutually externally tangent circles, how many circles are tangent to all three? This question is actually a specific case of Apollonius' problem back in Ancient Greece, which asks the same question for any three given circles in a plane. This article isn't about Apollonius, though, so let's stay with our mutually tangent circles. The answer is that there are exactly two circles, and these are named the Inner and Outer Soddy Circles. These two circles can be seen in Figure 1.

1.1 Brief History Lesson

Though these circles are called 'Soddy Circles', English Radiochemist Frederick Soddy did not discover them. As stated, these circles are a special case of Apollonius's problem, allegedly solved

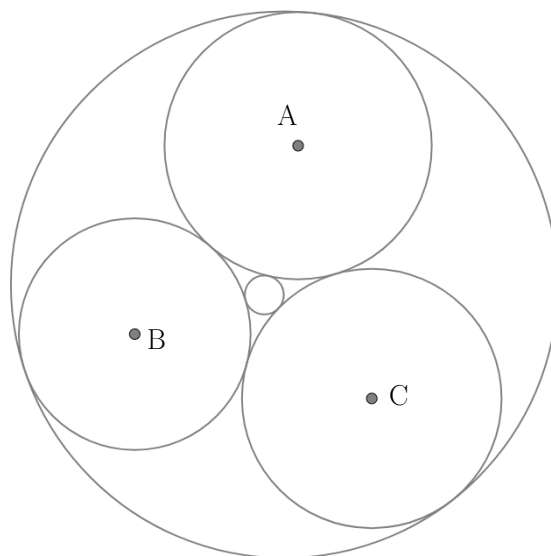


Figure 1: Inner and Outer Soddy Circles to Circles A , B , and C

in Apollonius of Perga's work *Tangencies* (English translation). The more formal statement of his problem is: Given three things (points, lines, or circles), construct a circle that passes through each point and is tangent to each line or circle. This text has been lost, though, and all that survives are references to the problems. Future mathematicians picked up the problem statement and went at solving it. This specific case of Apollonius's problem was picked up by René Descartes, who solved it in 1643 in a letter correspondence with Princess Elizabeth, the daughter of King Frederick of Bohemia. His formula gives the solution for the radii of both Soddy Circles, but the reasoning for his conclusion was not clear in the letter.

Descartes' Circle Theorem was rediscovered by multiple future mathematicians, one of whom being Frederick Soddy in 1936, who chose to publish his findings in the form of a poem, *The Kiss Precise*. This move by Soddy popularized the theorem in his name, which allowed Soddy to put his name on the circles. Soddy is actually better known for his Nobel Prize-winning work in chemistry as well as his work in Nuclear Physics and Economics. Descartes' Circle Theorem and its extensions were by no means discovered for the first time by Soddy. This is a common phenomenon in mathematics; it is very hard to find a problem that hasn't been explored before. This is not to say that Soddy just copied the theorem, he was a skilled mathematician and scientist and his contributions to these fields cannot be overlooked.

2 Existence / Construction

So how do we know there exist only two circles, and how do we know any exist at all? The tool we will use to prove the existence of the Soddy Circles is geometric inversion. This problem lends itself nicely to inversive geometry, due to the key property that inversion maintains tangency with respect to the reference circle.

2.1 Inversion

Inversion is the transformation of a figure with respect to a circle, which we call the reference circle. Let the reference circle have center O and radius r . Then, for any point P in the plane, we define the *inverse* of P with respect to the reference circle to be P' , such that P' is on the ray $\overrightarrow{OP}$ and

$$OP \cdot OP' = r^2.$$

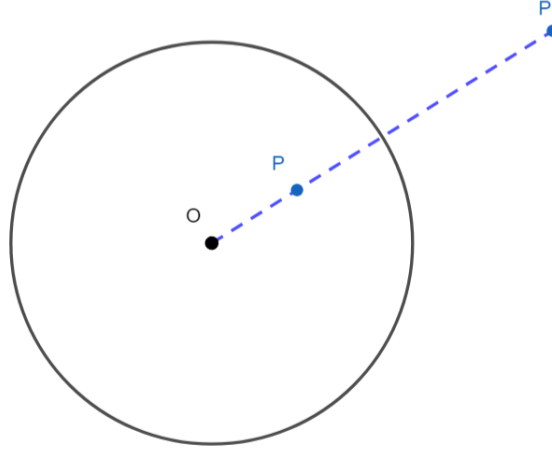


Figure 2: Visual representation of inversion of a point

From this definition we can see that the inverse of any point on the circumference of the reference circle is the same point. Considering Point S on the reference circle, we know the inverse S' is on $\overrightarrow{OS}$ and satisfies

$$OS \cdot OS' = r^2.$$

Since S is on the reference circle, $OS = r$. So, we have

$$\begin{aligned} OS \cdot OS' &= r^2 \\ r \cdot OS' &= r^2 \\ OS' &= r \end{aligned}$$

From this we know $S' = S$ since the only point on $\overrightarrow{OS}$ that is r away from O is point S . So, the inverse of any point on the circle is itself. Another way to say this is that the points on the reference circle are invariant under inversion. Similarly, we can see that the inverse of any point inside the reference circle lies outside the circle, and the inverse of any point outside the circle lies inside the circle.

2.1.1 Circles and Lines

With our definition of inversion, an interesting question to ask is: What is the inverse of the center of inversion? Let's try to find it using the definition! Let O' be the image of O under

inversion about the circle centered at O with radius r . Then,

$$\begin{aligned} OO \cdot OO' &= r^2 \\ 0 \cdot OO' &= r^2 \\ 0 &= r^2 \end{aligned}$$

There is seemingly no solution if we use the definition of inversion (given $r > 0$). It would be quite inconvenient to not be able to invert any figure that passes through the center of inversion. For any point P , we can express $OP' = \frac{r^2}{OP}$ where P' is the image of P . Then, we can make a limiting argument, assuming OO' exists, to get

$$\begin{aligned} \lim_{P' \rightarrow O'} OP' &= \lim_{P \rightarrow O} \frac{r^2}{OP} \\ OO' &= \lim_{P \rightarrow O} \frac{r^2}{OP} \\ OO' &= r^2 \lim_{P \rightarrow O} \frac{1}{OP} \end{aligned}$$

Validating the first step in this limiting argument likely requires the condition that inversion is a bijection, or that each point has exactly one inverse and that inverse is unique to that point. This is because of the limits we introduce to both sides, since we are saying that P corresponds to P' for every point as we approach O . Inversion is defined so that it feels slightly obvious that this is true, since there is always only one point on $\overrightarrow{OP}$ of any given positive length. This won't be proven here, but the proof would involve geometric continuity.

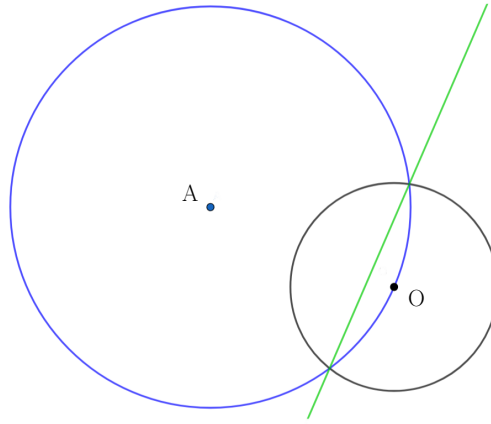


Figure 3: Line inverse of a circle passing through O

Since the length $OO = 0$, we know $\lim_{P \rightarrow O} \frac{1}{OP}$ is undefined (or, infinity). This means that OO' is also undefined. If we interpret this as the line segment OO' has infinite length, then we can say that O' is a *point at infinity*.

The basic figures that are used in inversive geometry are circles and lines (or, *clines*). Circles can invert to lines and vice versa. We have just shown how a circle can invert to a line: the inverse of a circle passing through the center of inversion is a line, since it must pass through a point at infinity! Proving that the inverted figure is exactly a line is more geometrically involved, so it will

remain unproven here (try it yourself!). When a circle inverts to a line, these lines are parallel to the tangent lines of the inverted circles at the center of inversion. In Figure 3, the green line is parallel to the tangent line to the blue circle through O .

Lines are sometimes considered to be degenerate circles with infinite radius, so they can be viewed as special cases of circles and be treated as such. With this interpretation, we can say that all circles are transformed to circles under geometric inversion.

2.1.2 Invariance of Orthogonal Circles

We know that the inverse of a circle intersecting the reference circle still intersects the reference circle, since the points of intersection lie on the reference circle and are thus invariant. When is the inverse of a circle invariant?

Along with points on the reference circle, orthogonal circles are preserved under inversion. Consider two circles, A and B , that intersect at two points. Circles A and B are called *orthogonal* if the tangent lines to A and B at each intersection point are perpendicular. This idea is important for our proof later. The most straightforward proof of this argument uses power of a point.

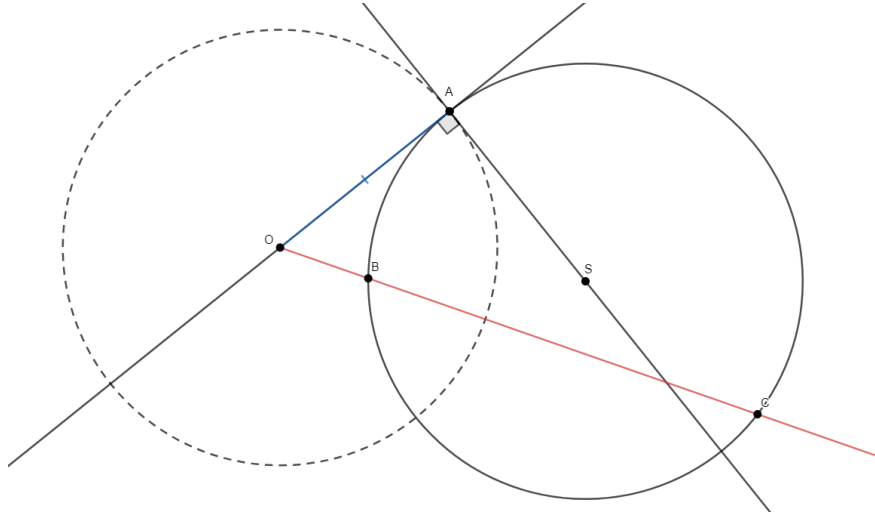


Figure 4: Orthogonal Circles O and S

Refer to Figure 4. Since the circles are orthogonal, the radius OA to the intersection point A is a tangent segment to circle S . We also consider an arbitrary ray $\overrightarrow{OB}$ through point B on circle S , and let point C be the other point that $\overrightarrow{OB}$ intersects S . We know OC is a secant, so we can use power of a point for point O with respect to circle S .

$$OA^2 = OB \cdot OC$$

Since A is on circle O , we know $OA = r$.

$$OB \cdot OC = r^2$$

Since this satisfies the definition of inversion, we have shown that B and C are inverses of each other with respect to circle O . So, any circle orthogonal to the reference circle will map onto itself under geometric inversion. The exception to this rule is if the orthogonal circle passes through O , in which then the circle inverts to a line.

2.1.3 Involutions, Bijections, and Other Inverses

When inverting a figure A to A' , we often call A' the *image* of A under inversion. Inversion is an *involution*, which means that the $(A')' = A$, or the inverse of the inverse of figure A is A . As mentioned before, we can see this pointwise.

Let P be the point we are inverting about circle O . Then, P' is the image of P and P'' is the image of P' . Note that, by definition, P' is on ray $\overrightarrow{OP}$, and P'' is on $\overrightarrow{OP'} = \overrightarrow{OP}$. We want to show that $P'' = P$. From the definition of inversion, we have

$$\begin{aligned}OP \cdot OP' &= r^2 \\OP' \cdot OP'' &= r^2\end{aligned}$$

Thus, we can express

$$OP = \frac{r^2}{OP'} = OP''$$

So, as long as OP' is defined, both P and P'' are the same distance from O . Since P and P'' lie on the same ray centered at O , we must have $P = P''$ since there is exactly one point on the ray satisfying $OP = x$ for any positive real x . This isn't a fully rigorous proof, but you should see the idea now that inversion is an involution.

In fact, knowing that inversion is an involution is enough to know that it is also a bijection (though showing inversion is a bijection directly is also possible). Consider a function $f : A \rightarrow A$ such that, for all $x \in A$, $f(f(x)) = x$, and try to show that this function must be bijective. To do this, try to individually show that this function must be injective and surjective from the given condition.

We can see the relation between geometric inversion and our notion of functional inverses or additive/multiplicative inverses. For example, the function $f(x) = \frac{1}{x}$ returns the multiplicative inverse of x , based on the relation $x \cdot \frac{1}{x} = 1$. We can also see the exception of 0 to this function corresponding to our exception of the center of inversion.

We have that these other notions of inverses are also involutions, since $f(f(x)) = x$ wherever $f(x)$ is defined. Taking the additive inverse as another example, we have the additive inverse of a is the unique value b such that $a + b = 0$. From this, we also know the inverse of b is a , since $a + b = 0$.

2.1.4 Inversion Exercises

Here are some exercises to test your new inversion skills. The last few are quite difficult, and you can find detailed solutions from Chapter 8 of Evan Chen's *Euclidean Geometry in Mathematical Olympiads* for the last three.

1. Find the inverse of the point $(4, 5)$ about the unit circle.
2. Find the inverse of the line $y = 4$ to the circle $x^2 + y^2 = 16$.
3. Circle O has radius 2. Circle B has radius 1 and is contained in circle O . What are the possible images of circle B under inversion about circle O ?

4. Circle A has diameter 1. Circles B , C , and D with diameter $\frac{1}{3}$ and are placed inside A , stacked directly vertically on another (in that order) such that every circle is tangent to two others. Perform an inversion with reference circle A , and then one with reference circle C . How can you relate these images, and what does that tell you about inversion as a transformation?
5. Circles A and B are externally tangent and have the same radius value. What is the image of circle A under inversion about circle B ? What about B about A ?
6. Consider an inversion centered at O . A and B are points outside the reference circle with images A' and B' . What can you figure out about triangles $\triangle OAB$ and $\triangle OA'B'$?
7. Given $\triangle ABC$, investigate what an inversion about the incircle does to points A , B , and C . Can you generalize this?
8. Congruent circles A , B , and C intersect at four points. Find the number of circles that exist tangent to any two of A , B , and C (and prove it!)
9. Try to prove Ptolemy's Theorem with inversion. Recall that Ptolemy's states that: If four coplanar points A, B, C , and D are concyclic, then

$$AB \cdot CD + BC \cdot AD = AC \cdot BD.$$

Try to prove the converse as well!

10. Let p be the semiperimeter of $\triangle ABC$. Points E and F are taken on line AB such that $CE = CF = p$. Prove the circumcircle of $\triangle EFC$ is tangent to the excircle of $\triangle ABC$ corresponding to side AB .
11. (2019 AIME II) Triangle ABC has side lengths $AB = 7$, $BC = 8$, and $CA = 9$. Circle ω_1 passes through B and is tangent to line AC at A . Circle ω_2 passes through C and is tangent to line AB at A . Let K be the intersection of circles ω_1 and ω_2 not equal to A . Then $AK = \frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.
12. (EGMO) Let $\triangle ABC$ have orthocenter H and altitudes $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$. Perform an inversion around C with radius $\sqrt{CH \cdot CF}$. Where do the six points go?
13. (EGMO) Let A, B, C be three collinear points (in that order) and construct three semicircles $\Gamma_{AC}, \Gamma_{AB}, \omega_0$, on the same side of AC , with diameters $\overline{AC}$, $\overline{AB}$, and $\overline{BC}$ respectively. For each positive integer k , let ω_k be the circle tangent to Γ_{AC}, Γ_{AB} , and ω_{k-1} . Let n be a positive integer. Prove that the distance from the center of ω_n to $\overline{AC}$ is n times its diameter.
14. (USAMO 2009/5) Trapezoid $ABCD$, with $\overline{AB} \parallel \overline{CD}$, is inscribed in circle ω and point G lies within $\triangle BCD$. Rays $\overrightarrow{AG}$ and $\overrightarrow{BG}$ meet ω again at points P and Q respectively. Let the line through G parallel to $\overline{AB}$ intersect $\overline{BD}$ and $\overline{BC}$ at points R and S , respectively. Prove that quadrilateral $PQRS$ is cyclic if and only if $\overline{BG}$ bisects $\angle CBD$.

2.2 Existence of Soddy Circles

After that brief overview of inversion, we are ready to prove the existence of the Soddy Circles. First, we have to show that it is possible to have three mutually externally tangent circles. Soddy Circles and centers are often referred to with respect to a triangle, and we will soon see why. Given

an arbitrary triangle ABC , we are able to construct three mutually externally tangent circles centered at the vertices of $\triangle ABC$. Do you see how?

Consider the incircle of $\triangle ABC$. Let the points of tangency between the incircle and $\triangle ABC$ be A_1, B_1, C_1 on BC, AC , and AB respectively, as in Figure 5. You may recognize this setup if you've seen a proof for Heron's formula. If two tangent segments to a circle originate from the same external point, then those two segments are congruent. From this, we have $AB_1 = AC_1$, $BA_1 = BC_1$, and $CA_1 = CB_1$. Defining $a = AB_1$, $b = BC_1$, and $c = CA_1$, we can construct circles centered at A, B , and C with radii a, b , and c respectively. These circles will be mutually tangent at A_1, B_1 , and C_1 , since we have guaranteed that each pair of circles intersecting at these points are tangent to each other.

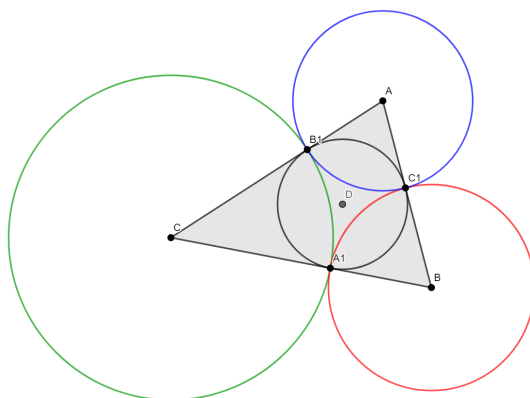


Figure 5: Construction of three mutually tangent circles

Now that we have constructed three mutually externally tangent circles, we can prove the existence of the Soddy Circles.

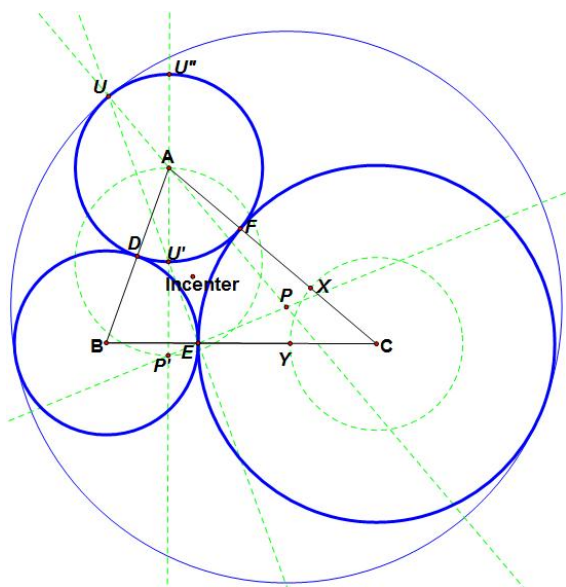
For the clever inversion, we choose the center of inversion to be A_1 (or any of the tangency points). We choose a radius such that the reference circle is orthogonal to circle A , so it is invariant under inversion about our reference circle.

Since both circles B and C pass through the center of inversion, A_1 , their images under inversion are lines. Both of their images are parallel to the tangent line to circles B and C through A_1 , so both lines are perpendicular to BC . We also know that, since both circles are tangent to circle A prior to inversion, they will also be tangent to A after inversion at the images of B_1 and C_1 .

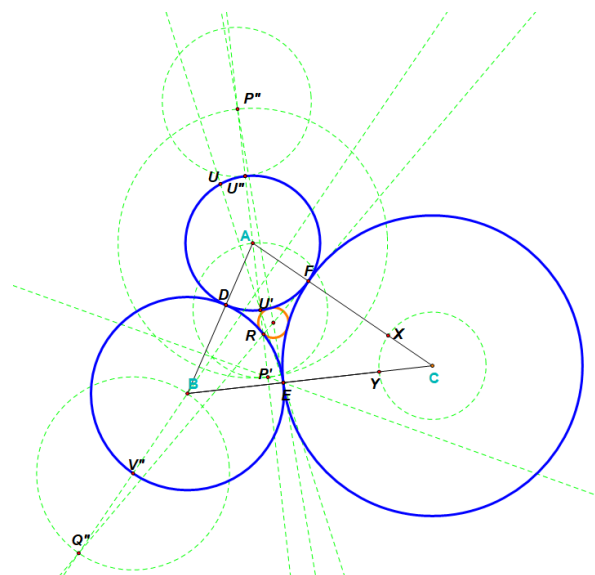
We have determined that the images of circles B and C under inversion are two lines tangent to circle A and perpendicular to BC . Let the points of tangency to circle A be T_1 and T_2 . We know T_1T_2 is parallel to BC since the lines are perpendicular to BC and tangent to circle A , so T_1T_2 (which passes through A) completes a rectangle between BC and the inverted lines.

Now, our Soddy Circles must be tangent to both of the inverted lines and circle A , since that's what we want pre-inversion. There are only two circles that satisfy this, as seen in Figure 6. These circles cannot be on the opposite side of either line as circle A , because then it cannot be tangent to A . So, there is a fixed radius for these circles as they are restricted between both lines. Then, only two circles that are contained within the lines are also tangent to circle A . So, we have found

7. Construct EU' . This segment intersects circle A at U .
8. Construct P' on line AU' such that $U'P' = AU'$ and $P' \neq A$.
9. Let P be intersection point of AU and EP' .
10. Construct a circle centered at P with radius UP . This is the outer Soddy Circle.



(a) Outer Soddy Circle Construction



(b) Inner Soddy Circle Construction

Figure 7: Soddy Circle Constructions (Krista Floer)

2.3.2 Inner Soddy Circle

Following the same steps as the Outer Soddy Circle, we have $\triangle ABC$ with AB as the shortest side. Refer to Figure 7b as you read the construction process.

1. Follow steps 1-5 from the Outer Soddy Circle Construction. Note that this is the construction the three tangent circles centered at the triangle vertices.
2. Construct a point U'' as the intersection of circle A and the line through A perpendicular to BC . This point is the intersection point farthest from side BC .
3. Construct point U as the other point of intersection between EU' and circle A .
4. Construct P' on line AU' such that $U'P' = AU'$ and $P' \neq A$.
5. Construct a circle centered at A with radius AP' . Point P'' is the intersection of this circle and line AU' .
6. Construct Point F as the point of tangency between circles A and C .
7. Construct point V'' as the intersection of the line through B perpendicular to AC . The intersection point farthest from side AC is V'' .

8. Construct a circle with center V'' and radius $V''B$. The intersection point of this circle and line $V''B$ is point Q'' .
9. Construct lines FQ'' and EP'' . Let the intersection of these two lines be S . This is the center of the inner Soddy circle.
10. Construct point R as the intersection of circle B and line FQ'' .
11. Construct a circle centered at S with radius RS . This is the inner Soddy Circle.

2.4 Uniqueness

How do we know that the Soddy Circles of a triangle are unique? That is, do we know that there are only two Soddy Circles for any given triangle? Well, we know that for any three mutually externally tangent circles, there exists exactly two circles tangent to both (which is what we proved with inversion). So, as long as a triangle uniquely determines three mutually externally tangent circles centered at its vertices, then the Soddy Circles are unique.

Our mutually tangent circles were formed based on the tangent points of the incircle to the triangle. If there was another triplet of circles, then there would have to be another corresponding incircle to the triangle. However, this cannot be true since the incircle of a triangle is unique. The incenter must be equidistant from all three sides of the triangle, so it is formed by the intersection of the angle bisectors of each angle in the triangle. The angle bisectors of a triangle are concurrent at a single point, the incenter. The incenter is unique, so the incircle must also be unique. Thus, Soddy Circles/centers are unique for any given triangle.

3 Descartes' Circle Theorem

Descartes' Circle Theorem was stated by Descartes in 1643 but left unproved. Since then, proofs have been discovered and the theorem has been extended. For any four mutually tangent circles with radii r_1, r_2, r_3 , and r_4 , Descartes' Circle Theorem states that

$$2\left(\frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} + \frac{1}{r_4^2}\right) = \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{r_4}\right)^2.$$

This statement can be simplified by taking $b_i = \frac{1}{r_i}$, getting

$$2(b_1^2 + b_2^2 + b_3^2 + b_4^2) = (b_1 + b_2 + b_3 + b_4)^2.$$

The quantity b_i is sometimes called the curvature or bend of a circle. We can use this equation to solve for the radii of the Inner and Outer Soddy Circles when given three mutually externally tangent circles. Note that, whenever solving, we will get a positive and a negative solution. Two solutions make sense, since there are exactly two circles tangent to all three circles.

The negative solution corresponds to the radius of the Outer Soddy Circle, for which we say it has negative radius/curvature since the other three circles are internally tangent to it. Similarly, the Inner Soddy Circle's radius corresponds to the positive solution of the quadratic, since the Inner Soddy Circle is externally tangent to all three circles.

3.1 Proof

Descartes' Circle Theorem can be proved using only Heron's formula and algebraic manipulation. We consider the line segments from the vertices of the triangles to the inner/outer Soddy centers. First, we know the sides of the triangle to be $r_1 + r_2$, $r_2 + r_3$, and $r_1 + r_3$, given these are

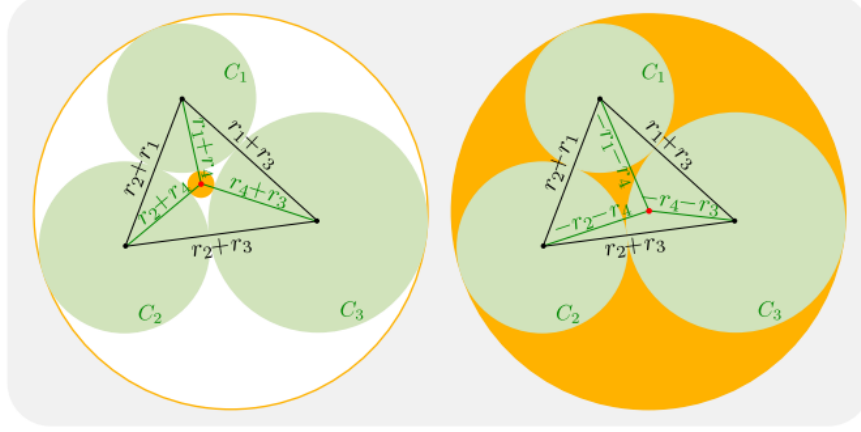


Figure 8: Labeled Diagrams for proof (Paul Levrie)

radii of the mutually externally tangent circles. Let side AB have length $r_1 + r_2$, side BC have length $r_2 + r_3$, and side AC have length $r_1 + r_3$.

For the inner Soddy center, we know the length of these segments to be $r_1 + r_4$, $r_2 + r_4$, and $r_3 + r_4$, where r_4 is the radius of the Inner Soddy Circle. Let the inner Soddy Center be O_1 , then $AO_1 = r_1 + r_4$, $BO_1 = r_2 + r_4$, and $CO_1 = r_3 + r_4$.

Since, when solving for radius using Descartes' Circle Theorem, the radius of the Outer Soddy Circle is considered to be negative, we consider $-R_4$ to be the radius of the Outer Soddy Circle with center O_2 . Then, we have that the segments have lengths $AO_2 = -R_4 - r_1$, $BO_2 = -R_4 - r_2$, and $CO_2 = -R_4 - r_3$. We will show the two cases to be equivalent when considering Heron's formula.

Heron's formula is a formula for triangle area. For $\triangle ABC$ with side lengths a, b, c , we define the semiperimeter to be $s = \frac{a+b+c}{2}$. Then,

$$[\triangle ABC] = \sqrt{s(s-a)(s-b)(s-c)}.$$

We know the side lengths of all of the triangles formed by the segments we constructed, so we can use Heron's formula.

First, we can find the area of the large triangle. The semiperimeter is

$$s = \frac{2r_1 + 2r_2 + 2r_3}{2} = r_1 + r_2 + r_3,$$

so the area via Heron's formula can be expressed as

$$[\triangle ABC] = \sqrt{(r_1 + r_2 + r_3)r_1r_2r_3}.$$

We know that, in both cases, the sum of the areas of the three triangles formed using the segments to the Soddy center is the large triangle.

For the inner Soddy center, we have three triangles. Then we have

$$\begin{aligned}s_{ABO_1} &= \frac{2r_1 + 2r_2 + 2r_4}{2} = r_1 + r_2 + r_4 \\s_{ACO_1} &= \frac{2r_1 + 2r_3 + 2r_4}{2} = r_1 + r_3 + r_4 \\s_{BCO_1} &= \frac{2r_2 + 2r_3 + 2r_4}{2} = r_2 + r_3 + r_4\end{aligned}$$

Similarly to $\triangle ABC$, we have the area of each subtriangle to be

$$\begin{aligned}[ABO_1] &= \sqrt{(r_1 + r_2 + r_4)r_1r_2r_4} \\[ACO_1] &= \sqrt{(r_1 + r_3 + r_4)r_1r_3r_4} \\[BCO_1] &= \sqrt{(r_2 + r_3 + r_4)r_2r_3r_4}\end{aligned}$$

Now, we can solve for the areas since $[ABC] = [ABO_1] + [ACO_1] + [BCO_1]$:

$$\sqrt{(r_1 + r_2 + r_3)r_1r_2r_3} = \sqrt{(r_1 + r_2 + r_4)r_1r_2r_4} + \sqrt{(r_1 + r_3 + r_4)r_1r_3r_4} + \sqrt{(r_2 + r_3 + r_4)r_2r_3r_4}.$$

From this step, only algebraic manipulation will get us Descartes' Circle Theorem. That won't be shown here, though, because it's very long.

We can perform the same steps for the Outer Soddy Circle, except all the semiperimeters and triangle sides will be negative. We get the same result, though, since there are four terms in the product for Heron's formula, so the factors of -1 will cancel out. So, we manipulate the above equation to get

$$2\left(\frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} + \frac{1}{r_4^2}\right) = \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{r_4}\right)^2,$$

which gives us the solution for both Soddy radii.

3.2 Degenerate Cases

As mentioned before, lines can be considered circles of infinite radius. It also makes sense, with our definition of curvature, that a line has 0 curvature. So, we have a few degenerate cases when some of the circles are lines.

3.2.1 One line

Since the curvature of a line is 0, this reduces Descartes' Theorem to

$$2(b_1^2 + b_2^2 + b_3^2) = (b_1 + b_2 + b_3)^2.$$

3.2.2 Two lines

Descartes' Theorem reduces to

$$2(b_1^2 + b_2^2) = (b_1 + b_2)^2.$$

In this case, we can solve for the radii and it isn't ugly. We can expand, getting

$$\begin{aligned} 2b_1^2 + 2b_2^2 &= b_1^2 + 2b_1b_2 + b_2^2 \\ b_1^2 + b_2^2 &= 2b_1b_2 \\ b_1^2 - 2b_1b_2 + b_2^2 &= 0 \\ (b_1 - b_2)^2 &= 0 \end{aligned}$$

Which we can see the only solution is $b_1 = b_2$, which is forced. We can see the relation to our inversion proof, where the the Soddy Circles under inversion had to be tangent to two lines and one circle, and they were both congruent to the circle.

3.2.3 Three lines

We have

$$2b_1^2 = b_1^2.$$

The only solution for this is when $b_1^2 = 0$, or $b_1 = 0$. This might not make sense, since we can have circles tangent to three different lines, but our lines are specifically degenerate circles. Since we require these 'circles' to be tangent, they must be parallel lines which intersect at infinity (remember how we inverted circles to lines). There cannot be a circle that is tangent to three distinct parallel lines, unless it is a parallel line (a degenerate circle) itself. This idea might seem confusing, but remember that all of the figures we are dealing with must be circles, even when considering lines as the degenerate case.

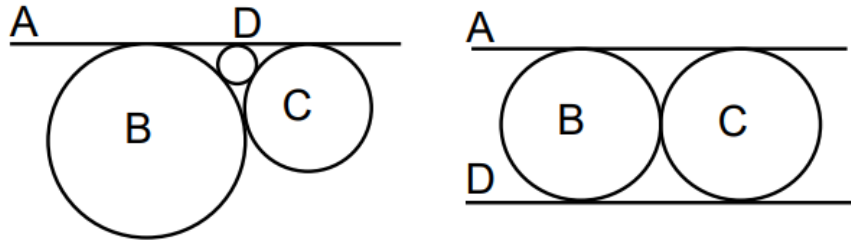


Figure 9: Two interesting degenerate cases of Descartes' Circle Theorem

4 Soddy Sequences

We can repeat the process of constructing the Inner Soddy Circle for any three mutually externally tangent circles to get a sequence of smaller and smaller Inner Soddy Circles. It is an interesting question to ask if, when repeating this infinitely many times, does the radius of the Soddy Circle converge? Well, every time we construct a new Soddy Circle, the area in which the next Soddy Circle can go (given we use the $(n - 1)^{\text{th}}$ Soddy Circle) is strictly decreasing. Since area must be positive, we know that the area that the next Soddy Circle can go approaches 0, so the radius of the n^{th} Soddy Circle approaches 0 as well.

4.1 Circle Packing

A cool application of this is circle packing, which is the study of figures of circles in a plane that all touch each other. When we take infinitely many Soddy Circles, the figure becomes a fractal, like in Figure 10.

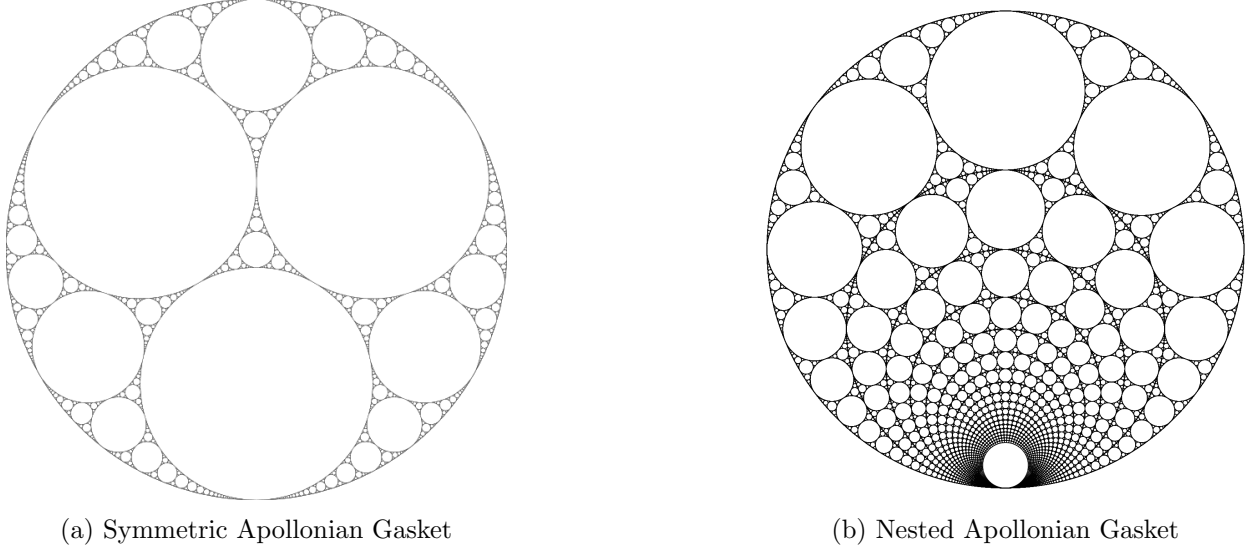


Figure 10: Fractals formed with infinitely many Soddy Circles

These packings are generated by an initial quadruple of circle curvatures, where three circles are mutually tangent within a larger one (like the Outer Soddy Circle). Then, the Inner Soddy Circle construction is applied indefinitely to get packed figures. Packings can also just be generated by triples, since we have shown that three mutually tangent circles determine the outer tangent one, but the outer circle is often included because packings only utilize the inner Soddy construction. These packings are often called Apollonian Gaskets after Apollonius. An interesting property of Apollonian Gaskets is that, if the initial quadruple of curvatures are all integers, then every circle in the gasket will also have an integer curvature. The proof of this utilizes Descartes' Circle Theorem, which we will use with curvatures:

$$2(b_1^2 + b_2^2 + b_3^2 + b_4^2) = (b_1 + b_2 + b_3 + b_4)^2.$$

Where $b_i = \frac{1}{r_i}$ for each circle. Let b_4 be the curvature we are solving for. Also, let $b_1 + b_2 + b_3 = n$ and $b_1b_2 + b_1b_3 + b_2b_3 = m$. Then, we have

$$\begin{aligned} 2(b_1^2 + b_2^2 + b_3^2 + b_4^2) &= (b_1 + b_2 + b_3 + b_4)^2 \\ 2((b_1 + b_2 + b_3)^2 - 2(b_1b_2 + b_1b_3 + b_2b_3) + b_4^2) &= (b_1 + b_2 + b_3 + b_4)^2 \\ 2(n^2 - 2m + b_4^2) &= (n + b_4)^2 \\ 2n^2 - 4m + 2b_4^2 &= n^2 + 2nb_4 + b_4^2 \\ b_4^2 - (2n)b_4 + n^2 - 4m &= 0 \end{aligned}$$

Now, we have a quadratic in b_4 , which we can solve with the quadratic formula:

$$\begin{aligned}
b_4 &= \frac{2n \pm \sqrt{4n^2 - 4(n^2 - 4m)}}{2} \\
&= \frac{2n \pm \sqrt{16m}}{2} \\
&= n \pm 2\sqrt{m} \\
&= b_1 + b_2 + b_3 \pm \sqrt{b_1b_2 + b_1b_3 + b_2b_3}
\end{aligned}$$

We have two solutions for b_4 . These are the curvatures of the inner and Outer Soddy Circles, so one is negative (outer) and the other is positive (inner). Let the Inner Soddy Circle have curvature $B = b_1 + b_2 + b_3 + \sqrt{b_1b_2 + b_1b_3 + b_2b_3}$ and the Outer Soddy Circle have curvature $B' = b_1 + b_2 + b_3 - \sqrt{b_1b_2 + b_1b_3 + b_2b_3}$. Then, $B + B' = 2(b_1 + b_2 + b_3)$, and we get

$$B = 2(b_1 + b_2 + b_3) - B'.$$

With the precondition that the starting quadruple (b_1, b_2, b_3, B') are all integers, we have shown that B will always be an integer. Due to this property, gaskets with integer starting quadruples, called Integral Apollonian Gaskets, are often studied.

4.2 Gasket Symmetry

We can construct these fractals to have as many symmetries as we want by changing the number of starting circles and having them be congruent. However, symmetry becomes much more restrictive when we only look at integral Apollonian circle packings. Integral packings can either have no symmetry, D_1 symmetry, or D_2 symmetry, where D refers to the dihedral symmetry group.

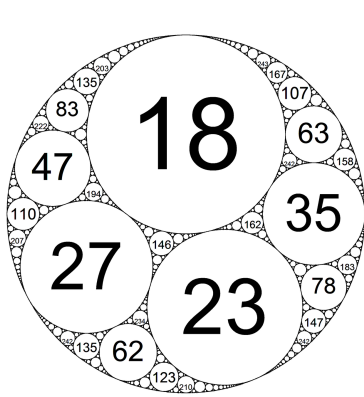
An integral Apollonian packing has no symmetry if the starting three curvatures (within the fourth circle) are all different. This makes sense, because there cannot be any rotational or reflectional symmetry if the three largest circles do not share values.

D_n symmetry refers to a group of transformations of rotations and reflections. For a figure to be D_n symmetric, it should be symmetric about n lines through the origin and by rotation of all multiples of $\frac{2\pi}{n}$ about the origin. Basically, it must have n different rotational symmetries (including 2π) and n different reflectional symmetries. Refer to figure 11 to see examples of each symmetry.

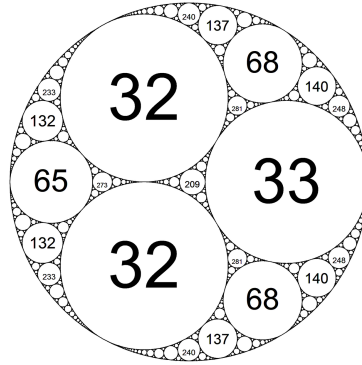
There are many integral packings with D_1 symmetry. The only requirement for this is that any two of the largest 5 circles have the same curvature. This gives us one line of reflectional symmetry about a diameter and no rotational symmetry in these packings.

There exists only one packing with D_2 symmetry: the packing with the first 5 curvatures $(-1, 2, 2, 3, 3)$. The -1 refers to the outer circle, but then there are two pairs of the same integer, so there are two lines of reflectional symmetry, and a rotational symmetry by π .

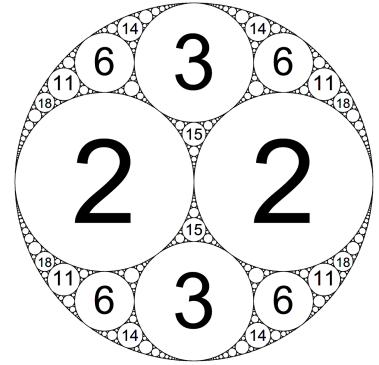
No packings exist with D_3 symmetry; This is because it would require three starting circles with the same curvatures. It can be shown that the ratio of the curvatures of these congruent circles to the circle they are inscribed in is irrational, so there exists no integral apollonian gasket with D_3 symmetry.



(a) No symmetry



(b) Close to D_3 , but only D_1 symmetry



(c) The only case of D_2 symmetry

Figure 11: Integral Apollonian Gaskets

4.3 Interesting Case

There is an interesting case of circle packing in the degenerate case of Descartes' Circle Theorem with two lines and two circles. The lines have curvature 0, and the pattern repeats infinitely within the lines, creating a stripe figure, as seen in Figure 12. This case corresponds to the starting quadruple $(0, 0, 1, 1)$. Each value of curvature repeats infinitely, but there remains only D_1 symmetry being the line equidistant from both parallel lines. We can show that every square of an

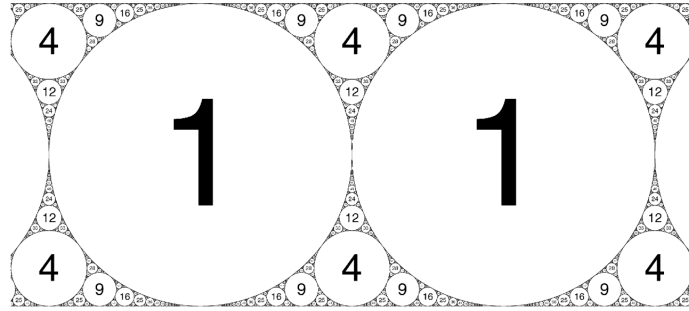


Figure 12: Strip Circle Packing

integer appears in the sequence of curvatures for this figure. Considering all of the circles tangent to one of the lines and a circle of radius 1, we have $B = 2(a + b + c) - B'$. For the first of these circles, it is tangent to two circles of curvature 1, so we have $B = 2(1 + 1 + 0) + 0 = 4$. We consider one of the lines to be the Outer Soddy Circle, the other figure tangent to the three circles we are considering. Then, the next circle in this sequence is tangent to a line, a circle with curvature 1, and a circle of curvature 4. This gets us $B = 2(1 + 4 + 0) - 1 = 9$. We consider the circle of curvature 1 to be the outer circle, since it is the other circle tangent to each of the circles we are considering. Now, we can prove that all squares appear in the sequence.

I claim that the circles with curvatures $(n + 1)^2$ and $(n - 1)^2$ will be the circles tangent to the circles with curvatures 0, 1, and n^2 .

The circle with curvature $(n - 1)^2$ will be the outer Soddy case, since a smaller curvature corresponds to a larger circle. We will show that, plugging in the values from the claim, we get an

identity that is always true:

$$\begin{aligned}
 B &= 2(1 + 0 + n^2) - (n - 1)^2 \\
 &= 2 + 2n^2 - n^2 + 2n - 1 \\
 &= n^2 + 2n + 1 \\
 &= (n + 1)^2.
 \end{aligned}$$

We have shown that $B = (n + 1)^2$ given the conditions, which validates the claim. So, if we have a circle with curvature $(n - 1)^2$ that is tangent to circles of curvature 0, 1, and n^2 , we have shown that the other tangent circle to these three circles has curvature $(n + 1)^2$. We have already shown the base case of this hypothesis, that a circle of curvature $4 = 2^2$ is tangent to the circles of curvature 0, 1, and 1, with the other solution being $(2 - 2)^2 = 0$. Then, we can use curvatures 0, 1, and 4 as the next case, which will get us a new circle of curvature 9, using the fact that we have already shown a circle of curvature 1 is the other solution to these constraints.

Researchers have questioned the patterns in the integers that appear in the sequences formed by integral Apollonian gaskets. In fact, questions about integral Apollonian gaskets are still being studied today. In the summer of 2023, a group of researchers proved the *Local-Global Conjecture* false. In any given primitive integral Apollonian circle packing (a packing is primitive if the starting quadruple are all relatively prime), each curvature value must belong to either six or eight residue classes modulo 24. The local-global conjecture stated that every integer part of either of these residue classes must appear as a curvature, and this was proven false with many counterexamples.

5 Multidimensional Soddy Hyperspheres

The Kiss Precise (Continued)

Frederick Soddy, 1936

*To spy out spherical affairs
 An oscular surveyor
 Might find the task laborious,
 The sphere is much the gayer,
 And now besides the pair of pairs
 A fifth sphere in the kissing shares.
 Yet, signs and zero as before,
 For each to kiss the other four,
 The square of the sum of all five bends
 Is thrice the sum of their squares.*

The Kiss Precise (Generalized)

Thorold Gosset, 1937

*And let us not confine our cares
 To simple circles, planes and spheres,
 But rise to hyper flats and bends*

*Where kissing multiple appears,
 In n-ic space the kissing pairs
 Are hyperspheres, and Truth declares -
 As $n + 2$ such osculate
 Each with an $n + 1$ fold mate
 The square of the sum of all the bends
 Is n times the sum of their squares.*

In Soddy's original poem, he detailed his findings that Descartes' Circle Theorem generalizes to spheres, in which five spheres can be mutually externally tangent and Descartes' Theorem becomes

$$3(b_1^2 + b_2^2 + b_3^2 + b_4^2 + b_5^2) = (b_1 + b_2 + b_3 + b_4 + b_5)^2.$$

Expressing this in summation notation, it becomes

$$3 \cdot \sum_{i=1}^5 b_i^2 = \left(\sum_{i=1}^5 b_i \right)^2.$$

In 1937, Thorold Gosset extended Soddy's poem with his own stanza, detailing how the *Soddy-Gosset* theorem states that Descartes' Circle theorem can be extended to n -dimensions, with it being possible for $n + 2$ hyperspheres to be mutually tangent. Generalized to n -dimensions, Descartes' Circle theorem becomes

$$n \cdot \sum_{i=1}^{n+2} b_i^2 = \left(\sum_{i=1}^{n+2} b_i \right)^2.$$

The proof for this, as one can assume, will not use the simple geometry we used for Descartes' Circle Theorem on a 2-dimensional plane. Instead, a better method of generalizing to higher dimensions is to use matrix equations, and that's one way to prove this generalization.

5.1 Soddy's Hexlet

On January 9th, 1937, Frederick Soddy published his findings regarding 'The Bowl of Integers and the Hexlet', which are 3-dimensional sphere packings that use Descartes' Circle Theorem. The theorem for Soddy's Hexlet is stated as follows: Given two spheres A and B , both externally tangent to each other and internally tangent to a circle C , we can construct a 'chain' of spheres tangent to A , B , C , and two other spheres in the chain. Once the chain is completed, there will always be 6 spheres in the chain.

Soddy presented a proof for this in his publication by applying Descartes' Circle Theorem. First, define δ as the curvature of the first sphere in the chain (which can vary as long as tangencies hold).

Using the Generalized Descartes' Circle Theorem to 3 dimensions, we have two real solutions for b_5 , the fifth sphere, in terms of δ and the curvatures of A , B , and C . Instead of two solutions corresponding to inner and outer solutions, in this case, since C is already the outer sphere, both solutions will be within C and tangent to each sphere.

Let the sum of the curvatures of A , B , and C be σ . Then, let the two values of b_5 be ϕ and ϕ' . Similarly to how we found that the solutions for b_4 sum to $2(b_1 + b_2 + b_3)$ in section 4.1, Soddy

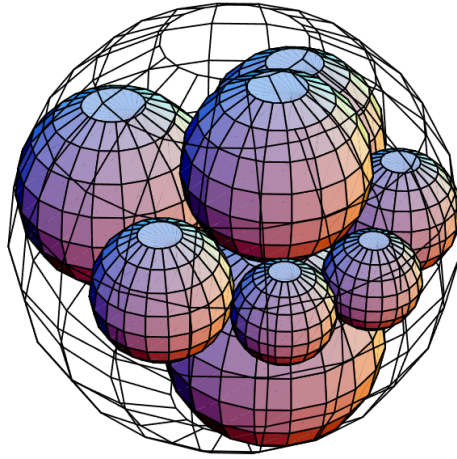


Figure 13: Soddy's Hexlet (Wolfram MathWorld)

found that, in 3 dimensions, the two solutions for curvatures also sum to a constant value. That is, we have the relation

$$\phi + \phi' = \sigma + \delta$$

We have guaranteed spheres with curvatures ϕ and ϕ' are tangent to the sphere with curvature δ by Descartes' Circle Theorem. So, ϕ, ϕ' , and δ are curvatures of consecutive spheres in the chain, and we can express this relationship recursively. Let ϕ_N be a the curvature of a sphere in the chain. Then, as Soddy found,

$$\phi_{N+1} + \phi_{N-1} = \sigma + \phi_N$$

From this, the curvatures of all spheres can be found (given any starting 2). Amazingly, this relation is periodic with period 6 (you can verify this yourself!), meaning every completed chain will consist of 6 spheres. This is why this figure is coined Soddy's Hexlet.

This theorem is actually a specific case of Kollros' theorem, which states that, for any ring of p spheres, there exists a ring of q spheres tangent to each of the p spheres such that

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{2}$$

In the case of the hexlet, we start with A, B , and C , so $p = 3$. We can easily see that $q = 6$ is the solution from Kollros' theorem.

5.2 Soddy's Bowl of Integers

In addition to the Hexlet, Soddy introduced the Bowl of Integers in his article. Beginning with a unit sphere, place two spheres of radius $\frac{1}{2}$ in it such that they are externally tangent to each other and internally tangent to the unit sphere. This configuration is called the 'Bowl of Integers', due to the property that the curvature of all spheres that can be packed in it are integers. This property bares a striking resemblance to the integral Apollonian circle packings, and that's because both rely on the Descartes' Circle Theorem. Since this is quite similar to what we did before, try to prove the relation yourself! The first few curvatures are $-1, 2, 5, 6, 9, 11$, and 14 .

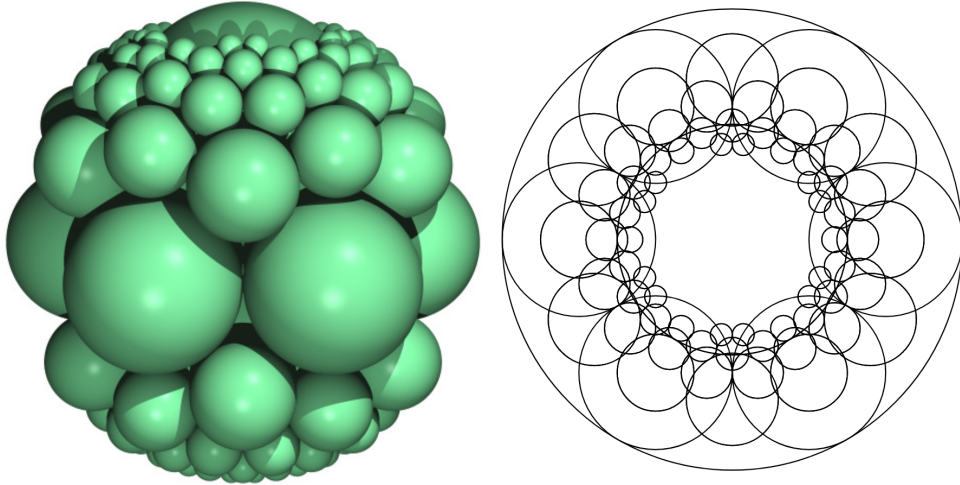


Figure 14: Bowl of Integers and 2 dimensional cross section

6 Non-Euclidean Spaces

The Kiss Precise (Further Generalized)

Fred Lunnon

*How frightfully pedestrian
My predecessors were
To pose in space Euclidean
Each fraternising sphere!
Let Gauss' k squared be positive
When space becomes elliptic,
And conversely turn negative
For spaces hyperbolic:
Squared sum of bends is sum times n
Of twice k squared plus squares of bends.*

Fred Lunnon extended Soddy's poem once more, describing a new generalization of Descartes' Theorem to non-Euclidean spaces, such as spherical, hyperbolic, and elliptical geometry.

6.1 Euclidean vs. Non-euclidean

Euclidean geometry is geometry on a flat plane, based on the axioms of Euclid in his book *Elements*. Non-Euclidean geometries are not conducted on planes, but rather on different 3D surfaces such as the surface of a sphere, ellipsoid, or hyperboloid. The main difference on these surfaces are in the notions of parallelism and perpendicularity of lines, which change fundamental properties that we are used to in Euclidean geometry. This article won't go in depth on non-Euclidean geometry, but Descartes' Circle Theorem generalizes quite nicely in these cases.

6.2 Descartes' Circle Theorem on Spheres and Hyperbola

In both spherical and hyperbolic geometry, curvature is defined differently than we have done so far in Euclidean geometry. In spherical geometry, a curvature $b_s = \cot(r_s)$ is defined in terms of r_s , named the *oriented intrinsic radius* of a circle, which is the radius of the circle on the surface of the sphere. Similarly, in hyperbolic geometry, a curvature $b_h = \coth(r_h)$, using the oriented intrinsic radius of a circle on a hyperbolic surface. Note that these two intrinsic radii have different definitions in relation to Euclidean geometry since we are relating different surfaces.

So, what does Descartes' Circle Theorem become? In spherical geometry, for three mutually tangent circles (curvatures b_1, b_2, b_3), the curvatures of the two circles tangent to all three (b_4) can be solved using the relation

$$(b_1 + b_2 + b_3 + b_4)^2 = 2(b_1^2 + b_2^2 + b_3^2 + b_4^2) + 4.$$

Notice that, this is exactly Descartes' Circle Theorem in Euclidean geometry with an added constant term! further, in hyperbolic geometry, we can solve for mutually tangent circle curvatures with the equation

$$(b_1 + b_2 + b_3 + b_4)^2 = 2(b_1^2 + b_2^2 + b_3^2 + b_4^2) - 4.$$

Interestingly, it is very similar to spherical geometry. In both cases, the fact we proved earlier still holds: for both solutions of b_4 (let those be B and B'), we have the relationship

$$B + B' = 2(b_1 + b_2 + b_3).$$

Which means we can create Soddy sequences and integral Apollonian circle packings on spherical and hyperbolic surfaces as well!

6.3 Putting it All Together

Combining the multidimensional and non-Euclidean generalizations of Descartes' Circle Theorem, we can get one Ultimate Descartes' Circle Theorem that applies in n -space and on different surfaces. In general n -space, we can guarantee $n + 2$ mutually tangent hyperspheres. Let the curvature of the i^{th} hypersphere be b_i . Then, we have

$$\left(\sum_{i=1}^{n+2} b_i\right)^2 = n \cdot \left(\sum_{i=1}^{n+2} b_i^2\right) + 2nv^2$$

Where v is the curvature of the space. For example, for hyperbolic geometry, $v^2 = -1$, in elliptical geometry, $v^2 = 1$, and in Euclidean geometry, $v^2 = 0$.

7 Conclusion

Taking a deep dive into mutually tangent circles in the lens of inversion allowed for a very straightforward proof of the existence of Soddy Circles. An associated result, Descartes' Circle Theorem, related the radii of these circles, and as it turns out the theorem generalizes far, far beyond two dimensional Euclidean geometry. Though there may not be much practical use to knowing one can pack infinitely many circles of integer curvature into a circle on a hyperbolic surface, Frederick Soddy and Descartes' Theorem are great examples of when mathematicians, driven purely by curiosity, aided by the findings of their predecessors, are able to explore problems

to their fullest extent. There appears to be a fundamental harmony present that allows us to generalize mutually tangent circles to the furthest extent with a simple equation. Should it just be chopped up to the ‘infinite symmetries’ of a circle? I don’t believe so, but questioning the structure that our mathematics has been built on is vital to advancing it. Is there some underlying basis that presents such a satisfying result? If there is, push it to its limits. If there isn’t, keep looking.

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